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civil engineering - 0 of 34 submitted

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24. T-187d4cc0

Item 24 of 34 - T-187d4cc0 - water resources

Your answer matches the template's answer within 1%.

You entered: 0.76 m^3/s

Question and solution

Source code

Question

During a design storm with rainfall intensity $i = 46$ mm/h, determine the peak discharge in m^3/s from a catchment with two land uses: 8.1 ha of single-family residential area ($C = 0.36$) and 4.2 ha of downtown business district ($C = 0.73$), the runoff coefficients taken from HEC-22 Table 3-1, using the rational method in SI form ($Q = CiA/360$ with i in mm/h and A in hectares).

Solution

Given: Rainfall intensity (i): 46 mm/h Catchment: a catchment with two land uses: 8.1 ha of single-family residential area ($C = 0.36$) and 4.2 ha of downtown business district ($C = 0.73$), the runoff coefficients taken from HEC-22 Table 3-1

Step 1: Compute the area-weighted composite runoff coefficient. $C_w = (C1A1 + C2A2) / (A1 + A2) = (0.36 * 8.1 + 0.73 * 4.2) / 12.3 = 0.486$

Step 2: State the rational formula in SI form. $Q = C_w * i * A / 360$, with Q in m^3/s , i in mm/h, and A in ha.

Step 3: Substitute and evaluate. $Q = 0.486 * 46 * 12.3 / 360 = 0.76 \text{ m}^3/\text{s}$

Answer: The peak discharge is $0.76 \text{ m}^3/\text{s}$

Another instance (1 of 5)

Your verdict

Physical plausibility

5

Mathematical correctness

Pedagogical clarity

5

5

Decision

☒ Approve ☐ Reject

Defect type (required on Reject)

Choose options

Note (required on Reject: what is wrong and where; optional on Approve)

Your confidence in this verdict

4

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